# Lecture 25 – Induction

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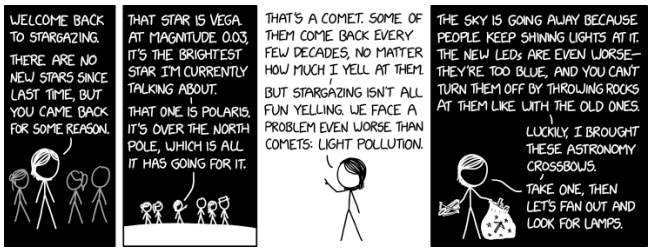

# 1 Faraday's Law

$$\oint \vec{E}.d\vec{S} = -\frac{d\phi}{dt}B$$

- Integral over an amperean loop
- $\frac{d\phi}{dt}$ is the rate of change of magnetic flux
- The negative (and possible positive) is indicative of direction. Lenz' law is used to decide this.